

Week 9 In-Class Worksheet

Correlation and Simple Linear Regression

Name: _____

Date: _____

Course: H524 Introduction to Biostatistics

Topic: Correlation, Regression, and Prediction

Instructions

Work with a partner to solve the following problems. Show all your work and calculations. You may use R to verify answers, but demonstrate your understanding by showing the steps.

1 Problem 1: Correlation Analysis

A researcher studies the relationship between daily calcium intake (mg) and bone density (g/cm^2) in 8 postmenopausal women:

Subject	Calcium (mg)	Bone Density (g/cm^2)
1	800	0.85
2	1000	0.90
3	1200	0.95
4	900	0.88
5	1100	0.92
6	1300	0.98
7	950	0.89
8	1050	0.91

Part A: Create a scatter plot of bone density (Y) vs. calcium intake (X). Describe the relationship you observe.

Part B: Calculate summary statistics:

Mean calcium intake: $\bar{X} =$ _____

Mean bone density: $\bar{Y} =$ _____

SD of calcium: $s_X =$ _____

SD of bone density: $s_Y =$ _____

Part C: The Pearson correlation coefficient is $r = 0.94$. Interpret this value in context.

Part D: Test whether the correlation is statistically significant at $\alpha = 0.05$.

Hypotheses:

H_0 : _____

H_A : _____

Test statistic: $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$

Degrees of freedom: $df =$ _____

Critical value (from t-table): $t_{0.975, df} =$ _____

Decision: _____

Conclusion (in context):

Part E: Calculate the coefficient of determination, r^2 . Interpret this value.

$r^2 =$ _____

Interpretation:

2 Problem 2: Simple Linear Regression

Using the same data from Problem 1, we want to predict bone density from calcium intake.

Part A: Calculate the regression slope using the formula:

$$b_1 = r \times \frac{s_Y}{s_X}$$

Given: $r = 0.94$, $s_Y = 0.042$ g/cm², $s_X = 167.7$ mg

Part B: Calculate the regression intercept using:

$$b_0 = \bar{Y} - b_1 \bar{X}$$

Given: $\bar{Y} = 0.910$ g/cm², $\bar{X} = 1037.5$ mg

Part C: Write the regression equation:

Bone $\widehat{\text{Density}} =$

Part D: Interpret the slope in context. What does it mean biologically?

Part E: Is the intercept meaningful in this context? Why or why not?

3 Problem 3: Predictions and Residuals

Continue with the regression model from Problem 2.

Part A: Use your regression equation to predict bone density for a woman with calcium intake of 1100 mg/day.

Calculation:

Predicted bone density: _____

Part B: Subject 5 from the original data has calcium intake 1100 mg and observed bone density 0.92 g/cm^2 . Calculate the residual for this subject.

Residual = Observed – Predicted = _____

Part C: Is this residual positive or negative? What does that tell you about this subject?

Part D: Would you feel confident predicting bone density for someone with calcium intake of 2000 mg/day? Why or why not?

4 Problem 4: Blood Pressure and Age

A study of 12 adults examines the relationship between age and systolic blood pressure. The regression output is:

	Estimate	Std. Error	t value	p-value
Intercept	80.5	8.2	9.82	< 0.001
Age	1.15	0.18	6.39	< 0.001

Residual standard error: 6.3 mmHg on 10 degrees of freedom
Multiple R-squared: 0.803, Adjusted R-squared: 0.783

Part A: Write the regression equation.

Part B: Interpret the slope coefficient in the context of this study.

Part C: Test whether there is a significant relationship between age and blood pressure at $\alpha = 0.05$.

Hypotheses:

H_0 : _____

H_A : _____

Test statistic: $t =$ _____ (from table above)

p -value: _____

Decision and conclusion:

Part D: Construct a 95% confidence interval for the slope β_1 .

Formula: $b_1 \pm t_{0.975,10} \times \text{SE}(b_1)$

Given: $t_{0.975,10} = 2.228$, $\text{SE}(b_1) = 0.18$

95% CI: _____

Interpretation (in context):

Part E: What percentage of the variability in systolic blood pressure is explained by age?

Part F: Use the regression equation to predict systolic blood pressure for a 50-year-old.

Part G: The residual standard error is 6.3 mmHg. What does this tell us about the model's predictions?

5 Problem 5: Comparing Methods

A pharmaceutical company studies the relationship between drug dosage (mg) and reduction in pain score (0-10 scale) for 15 patients. Summary statistics:

- $\bar{X} = 50$ mg, $s_X = 15$ mg
- $\bar{Y} = 4.2$ points, $s_Y = 1.8$ points
- Pearson correlation: $r = 0.72$

Part A: Calculate the regression slope and intercept.

$$b_1 = \underline{\hspace{4cm}}$$

$$b_0 = \underline{\hspace{4cm}}$$

Regression equation: $\underline{\hspace{4cm}}$

Part B: Calculate R^2 . How much of the variability in pain reduction is explained by dosage?

Part C: Predict the pain reduction for a dosage of 60 mg.

Part D: If you wanted to achieve an average pain reduction of 5 points, what dosage would you recommend? (Hint: Solve the regression equation for X when $\hat{Y} = 5$)

Reflection Questions

Answer these questions to deepen your understanding:

1. What is the difference between correlation and regression? When would you use each?
2. Explain the difference between r and r^2 . Which is more directly interpretable and why?
3. Why is it important to examine a scatter plot before calculating correlation or fitting regression?
4. A study finds a strong positive correlation ($r = 0.85$) between ice cream sales and drowning deaths. Does this mean ice cream causes drowning? Explain.
5. What is extrapolation, and why is it risky in regression analysis?

R Code Reference

Here are the R commands you can use to check your work:

For Problem 1:

```
# Enter data
calcium <- c(800, 1000, 1200, 900, 1100, 1300, 950, 1050)
bone_density <- c(0.85, 0.90, 0.95, 0.88, 0.92, 0.98, 0.89, 0.91)

# Summary statistics
mean(calcium)
sd(calcium)
mean(bone_density)
sd(bone_density)

# Correlation test
cor.test(calcium, bone_density)

# Scatter plot
plot(calcium, bone_density,
     main = "Bone Density vs. Calcium Intake",
     xlab = "Calcium (mg)",
     ylab = "Bone Density (g/cm2)",
     pch = 19)
```

For Problems 2-3:

```
# Fit regression model
model <- lm(bone_density ~ calcium)

# View results
summary(model)

# Extract coefficients
coef(model)

# Add regression line to plot
abline(model, col = "red", lwd = 2)

# Make predictions
predict(model, newdata = data.frame(calcium = 1100))

# Calculate residuals
residuals(model)
```

Checking assumptions:

```
# Diagnostic plots
par(mfrow = c(2, 2))
plot(model)
par(mfrow = c(1, 1))
```